

Layered World-Model Profiles:

Predictive Closure, Interventional Validity, Decision Sufficiency, and Robustness under Shift

Candidate Framework Spine v0.4

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Abstract

We propose a layered evaluation profile for systems commonly described as world models, while keeping open the eliminativist alternative of reporting the underlying technical tuple directly. Independent review of v0.3 reproduced both finite separations and cleared the executable-evidence blocker, but proved that the proposed behaviour-policy invariance quantity is bounded by the L0 aliasing quantity and therefore is not an independent coordinate. v0.4 demotes it to an attribution ratio inside L0, retains interventional validity as the independent L1 coordinate, repairs shrinking-support audit semantics, and narrows the empirical claim to finite non-identity of L0 and L2. Toy A has exact predictive closure at a phase abstraction yet positive contingent-decision regret; Toy B passes exact value equivalence on a declared value class while failing full-state predictive closure. No claim is made that these abstraction phenomena are novel, that L0–L3 is necessary and sufficient, or that non-collinearity has been established in a non-degenerate regime.

Epistemic status. This is a candidate framework spine after two independent review rounds. The current gate is whether the spine is coherent enough to justify expansion into a full arXiv manuscript, not whether it already constitutes such a manuscript.

1 From a binary predicate to a measured profile

World-model discussions often mix representation, prediction, action, planning, and robustness under one binary label. This creates the temptation to let a word such as “sufficient” carry most of the discrimination. We instead report a declared profile

$$\text{WMPProfile}_{\mathcal{D}}(X) = \langle L0[\rho_{\text{pol}}], L1_{\text{int}}, L2.G, L2.O, L3 \rangle.$$

Behaviour-policy exposure is no longer a separate axis: review of v0.3 showed that its population quantity is structurally bounded by L0 aliasing.

Let $h_t = (o_{\leq t}, a_{< t})$ and $z_t = \sigma(h_t)$. Before evaluation, declare

$$\mathcal{D} = \left(\underbrace{\mathbb{A}, H_{\mathbb{A}}, \Omega_0, \mathcal{F}, \mathcal{B}}_{L0}; \underbrace{\mathcal{I}}_{L1_{\text{int}}}; \underbrace{\mathcal{T}, H, \Pi, \pi_{\text{ref}}, \mathcal{U}, \varepsilon, R, \mathcal{N}}_{L2}; \underbrace{\Delta, \pi_{\text{ref}}}_{L3} \right).$$

Statuses are *pass with budget*, *fail*, *not declared*, or *not applicable*.

2 L0: predictive closure at an abstraction frontier

Define

$$P_{\sigma}(\cdot \mid h, a) = \text{Law}(\sigma(H_{t+1}) \mid H_t = h, A_t = a)$$

and a candidate latent kernel $K(\cdot \mid z, a)$.

2.1 Aliasing floor and kernel fit

For histories collapsed to the same latent,

$$\delta_{\text{alias}} = \sup_{\substack{h, h' \in \Omega_0 \\ \sigma(h) = \sigma(h')}} \sup_a d_{\mathcal{F}}(P_{\sigma}(\cdot \mid h, a), P_{\sigma}(\cdot \mid h', a)).$$

Worst-case kernel fit is

$$\delta_{\text{fit}} = \sup_{h \in \Omega_0} \sup_a d_{\mathcal{F}}(K(\cdot \mid \sigma(h), a), P_{\sigma}(\cdot \mid h, a)).$$

For the declared sup/IPM form,

$$\delta_{\text{alias}} \leq 2\delta_{\text{fit}}, \quad \frac{\delta_{\text{alias}}}{2} \leq \inf_K \delta_{\text{fit}}(K) \leq \delta_{\text{alias}}.$$

Thus δ_{alias} is best interpreted as an irreducible compression floor rather than an independent success criterion.

This coordinate is deliberately positioned against classical state-abstraction and bisimulation work rather than presented as a new abstraction theory. Stochastic bisimulation and model minimisation are developed by Givan et al. [1]; Li et al. [3] organise a hierarchy of model-, value-, action-, and policy-oriented abstractions; Ferns et al. [2] develop bisimulation metrics and value-related guarantees.

2.2 Conditional rollout and audit semantics

For a legal action sequence $a_{t:t+k-1}$,

$$\delta_{\text{roll}}(k) = \sup d_{\mathcal{F}}(K^k(\cdot \mid \sigma(h), a_{t:t+k-1}), P_{\sigma}^k(\cdot \mid h, a_{t:t+k-1})).$$

Every horizon-indexed report carries its admissible case count. An empty set is `NOT_APPLICABLE`; it is never converted to a zero error.

For each query $q \in \mathbb{A}$,

$$\eta_q(k) = \inf_{g_q} \sup d_q(\text{Law}(g_q(\hat{Z}_{t+k})), \text{Law}(q(S_{t+k}) \mid h, a_{t:t+k-1})), \quad 0 \leq k \leq H_{\mathbb{A}}.$$

In addition to total case count, a query-bearing count is reported whenever relevant histories can terminate or leave support. A numerical zero with no query-bearing histories is marked `NOT_INFORMATIVE`, not interpreted as improved decodability.

The resulting L0 claim is a frontier:

$$\mathbb{A}^* = \text{finest tested abstraction level that passes,}$$

together with a natural richer level that fails when available. This frontier is the main profile-level object presently added beyond a flat pass/fail predicate.

Self-predictive representation work provides a related modern representation perspective [8], while Approximate Information State supplies a distinct decision-planning framework in partially observed systems [4].

3 Policy exposure of L0 aliasing

For behaviour policy β , define the latent-conditioned mixture

$$K_{\beta}^{\sigma}(\cdot \mid z, a) = \int P_{\sigma}(\cdot \mid h, a) dP_{\beta}(h \mid z, a).$$

The v0.3 reviewer observed that these are mixtures of the member-history projected laws. By convexity of an IPM in the mixture weights,

$$0 \leq \delta_{\text{inv}}^{\text{latent}} \leq \delta_{\text{alias}},$$

with equality when the behaviour-policy class can concentrate on each member history.

Therefore $\delta_{\text{inv}}^{\text{latent}}$ is not retained as an independent coordinate. When $\delta_{\text{alias}} > 0$ and $\mathcal{B}$ is a declared restricted class, report instead

$$\rho_{\text{pol}} = \frac{\delta_{\text{inv}}^{\text{latent}}}{\delta_{\text{alias}}} \in [0, 1].$$

The interpretation is attribution: what fraction of existing compression aliasing can a change in the declared acquisition/behaviour class expose? If $\delta_{\text{alias}} = 0$, ρ_{pol} is NOT APPLICABLE. No novelty claim is made for this elementary mixture bound; literature prior-statement checking remains a manuscript task.

A finite witness uses two histories with projected laws δ_A and δ_B . Under the L1-TV convention, $\delta_{\text{alias}} = 2$. Restricted mixture weights $\{0.25, 0.75\}$ yield $\delta_{\text{inv}}^{\text{latent}} = 1$ and $\rho_{\text{pol}} = 0.5$; a rich class containing the vertices $\{0, 1\}$ yields $\rho_{\text{pol}} = 1$.

4 L1: interventional validity

Only interventional validity remains as the independent L1 coordinate. An abstract intervention claim is asserted only when σ is compatible with a declared causal-abstraction/exact-transformation relation. This guard is motivated by work on causal consistency and abstraction across model levels [10, 11].

When well posed,

$$\delta_{\text{do}} = \sup_{z, a} d_{\mathcal{F}}(K(\cdot \mid z, a), \text{Law}(\sigma(H_{t+1}) \mid z_t = z, \text{do}(A_t = a))).$$

In an unconfounded online setting where the action is the sole intervention point, this can collapse to ordinary transition fit. Additional empirical content arises under confounded offline data or interventions beyond the agent action. v0.4 does not yet include a dedicated $L1_{\text{int}}$ separation.

5 L2: decision-relative sufficiency

L2 is non-constitutive and relative to a declared decision problem.

AIS asks when a history statistic supports approximate dynamic programming through reward and transition-prediction conditions [4]. With matched metric/kernel choices, the L0 kernel-fit component corresponds to the transition-prediction side; reward prediction and theorem assumptions remain separate.

Value Equivalence instead asks which Bellman computations are preserved by a model over declared policy/function classes [5]; proper and approximate variants refine that programme [6, 7]. AIS and VE are therefore related adequacy routes but quantify different objects.

We separate class-level adequacy (L2.G) from operational model+search adequacy (L2.O). The split is motivated by evidence that MuZero’s learned model can be much more accurate in the regions favoured by its search/policy prior than for unseen policies [9]. An operational report under query distribution μ_{Π} must therefore include either

$$\sup_{\nu \in \mathcal{N}} E_{\nu}[\ell]$$

or a declared coverage/concentrability coefficient. Regret always names the comparator π_{ref} .

6 L3: robustness under a shift class

L3 is system-level:

$$\sup_{\delta \in \Delta} \text{Regret}_{\delta}(\pi; \pi_{\text{ref}}) \leq \rho.$$

It does not imply robustness of one particular latent kernel, because adaptation, memory, planning, or policy switching may compensate.

Richens and Everitt connect robustness to sufficiently broad local interventions with extractable causal structure under their stated conditions [12]. Related general-agent results show that predictive environment models can be extractable from sufficiently general agent behaviour [13]. These results must not be rewritten as claims of explicit internal SCM storage.

7 Finite evidence: exactly what is established

7.1 Toy A: L0 passes at one frontier, L2 contingent adequacy fails

Let hidden mode $M \in \{0, 1\}$ be uniform. An agent can commit immediately or pay cost c to probe and reveal M , then commit. The coarse representation retains only protocol phase.

At

$$\mathbb{A}_{\text{phase}} = \{q_{\text{phase}}\},$$

the exact finite computation gives zero aliasing, zero kernel-fit error, zero rollout error on admissible horizons, and zero phase-query error.

At the richer

$$\mathbb{A}' = \{q_{\text{phase}}, q_{\text{mode}}\},$$

mode-query error is 1 at $k = 0$ and $k = 1$ under L1-TV. At $k = 2$, the raw audit error is zero but there are no mode-bearing starting histories left, so the status is `NOT_INFORMATIVE`; at $k = 3$ the case set is empty and status is `NOT_APPLICABLE`.

Direct enumeration of all 12 history-measurable and 6 coarse policies gives

$$V_{\text{hist}}^*(c) = \max(\frac{1}{2}, 1 - c), \quad V_C^*(c) = \frac{1}{2},$$

hence

$$\text{Regret}(c) = \max(0, \frac{1}{2} - c).$$

At $c = 0.10$, regret is 0.40. L2 failure is certified directly by regret rather than inferred from failure of an AIS sufficient-condition bound.

7.2 Toy B: L2.G-VE passes while L0 fails

Let $s = (x, n)$, where x is decision-relevant and $n \in \{0, 1, 2, 3\}$ is nuisance noise. The true process draws n' uniformly; an intentionally wrong model always predicts $n' = 0$ while preserving the x' dynamics exactly. Reward depends only on x .

At full-state abstraction, the model-fit error is

$$\delta_{\text{fit}} = 1.5$$

under L1-TV. But for the declared value class

$$\mathcal{V}_x = \{V : V(x, n) = v(x)\},$$

the Bellman gap is exactly zero for all 256 deterministic stationary policies. This is intentionally close to classical noisy-TV/state-abstraction examples and is not claimed novel.

Together the examples establish only

$$L0 \not\Leftarrow L2, \quad L2.G \not\Leftarrow L0$$

in these finite constructions.

Scope limitation. They establish *non-identity*, not non-collinearity in a non-degenerate regime. Toy A uses a very coarse phase abstraction, and Toy B declares a value class blind to nuisance by construction. A stronger manuscript claim would require at least one non-degenerate separation.

8 Eliminativist alternative and MMALS boundary

The eliminativist alternative remains live: report

$$(\sigma, K, \text{sufficiency class}, \text{shift class}, \dots)$$

and retire “world model” as a technical predicate. The current profile demonstrably organises one extra object—the abstraction frontier ($\mathbb{A}^*$ passes, $\mathbb{A}'$ fails)—plus the ρ_{pol} attribution of exposed aliasing. That is real but thin, and v0.4 does not claim it defeats the flat tuple.

Adaptive abstraction revision,

$$\mathbb{A}_t \rightarrow \mathbb{A}_{t+1},$$

remains outside the spine. It is a prospective bridge to MMALS-style REUSE/ADAPT/FORK/NEW-REGIME logic rather than a hidden assumption of the current framework.

9 Conclusion

Independent review has progressively reduced the proposal rather than protecting it. v0.4 no longer treats behaviour-policy invariance as an independent axis, no longer allows shrinking support to masquerade as improved horizon performance, and narrows the empirical conclusion to finite L0/L2 non-identity. The next gate is whether this smaller profile—predictive abstraction frontier, interventional validity, decision sufficiency, and shift robustness—is coherent and useful enough to justify a full arXiv manuscript.

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